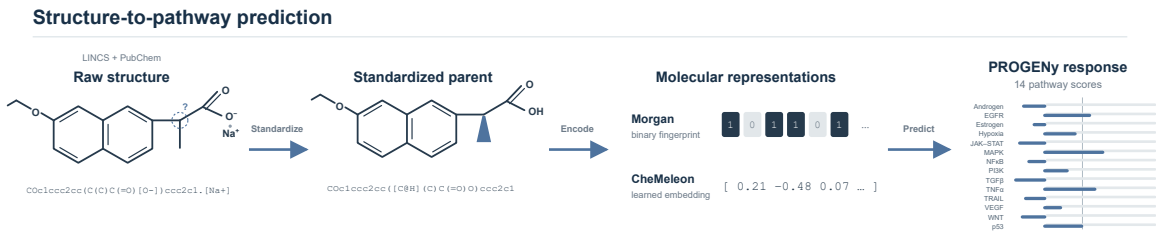


Predicting compound-induced pathway responses from molecular structure in human lung carcinoma cells

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Abstract

Predicting drug-induced biological responses directly from molecular structure could help prioritize compounds before experimental screening. We investigated whether molecular structure can predict transcriptionally inferred signaling-pathway responses in A549 human lung adenocarcinoma cells following compound exposure at 10 μ M for 24 hours. LINCS L1000 gene-expression signatures were converted into activity scores for 14 pathways using PROGENy, followed by adjustment for experimental block effects and aggregation to structure-level responses. The resulting dataset comprised 1,494 distinct molecular structures. We compared Morgan fingerprints and pretrained CheMeleon molecular embeddings across linear and nonlinear regression methods, including ridge regression, kernel ridge regression, support-vector regression, and a feedforward neural network. The best-performing method combined CheMeleon embeddings with pathway-specific radial basis function kernel ridge regression, achieving a mean test R^2 of 0.150 across the 14 pathways. Other nonlinear models produced similar but slightly lower performance, while Morgan and CheMeleon ridge regression achieved mean test R^2 values of 0.079 and 0.052, respectively. These findings show that molecular structure contains detectable information about compound-induced pathway responses, but the evaluated representations and models explain only a modest proportion of response variability. Accurate prediction may require additional information describing compound-cell interactions, cellular context, or experimental conditions. These conclusions are specific to pathway responses inferred in A549 cells under the studied dose and exposure duration.



Introduction

Lung adenocarcinoma is a major histological subtype of lung cancer and is characterized by extensive molecular heterogeneity. Genomic alterations, epigenetic dysregulation, and abnormal signaling networks can modify transcriptional programs that regulate cell proliferation, survival, metabolism, and responses to cellular stress (The Cancer Genome Atlas Research Network 2014). These changes in gene expression are therefore both consequences of oncogenic processes and measurable indicators of the biological state of a cancer cell. The A549 cell line, established from a human lung carcinoma, is widely used as an experimental model for studying lung cancer biology and pharmacological responses (Giard et al. 1973; ATCC, n.d.). Although a single cell line cannot represent the full heterogeneity of lung cancer, it provides a controlled cellular context in which compound-induced molecular responses can be compared.

Small-molecule compounds influence cellular behavior by interacting with molecular targets and altering downstream signaling networks. These effects can subsequently produce coordinated changes in gene expression. Measuring transcriptional responses after compound exposure therefore provides a means of investigating the cellular consequences of chemical perturbations. The Library of Integrated Network-based Cellular Signatures and Connectivity Map project generated large collections of standardized gene-expression profiles following chemical and genetic perturbations across multiple human cell lines, including A549 (Subramanian et al. 2017). These data make it possible to study systematic relationships between chemical structure and transcriptional response at a scale that would be difficult to achieve through individual experiments.

Gene-level expression profiles are high dimensional and can be difficult to interpret directly in terms of biological mechanisms. Pathway-level summaries provide a lower-dimensional representation of coordinated cellular responses. PROGENy infers signaling-pathway activity from downstream pathway-responsive genes rather than relying only on the expression of genes that belong to a pathway (Schubert et al. 2018; Saez Lab, n.d.). Consequently, PROGENy scores represent transcriptionally inferred pathway activity. In the present study, LINC1000 expression signatures measured in A549 cells after exposure to compounds at 10 μ M for 24

hours were summarized into activity scores for 14 pathways. These pathway scores provide the biological response variables used in the structure-to-pathway prediction problem.

The premise of structure-activity relationship modeling is that the chemical structure of a compound contains information about its biological properties. Molecular structure determines features such as shape, functional groups, electronic characteristics, and stereochemistry, which influence target binding, cellular exposure, and downstream biological effects. Quantitative structure-activity relationship models use numerical representations of these features to predict the activities of previously unobserved compounds (Cherkasov et al. 2014). Modern molecular representations range from substructure-based fingerprints to embeddings learned by pretrained graph neural networks. In this study, Morgan fingerprints and pretrained CheMeleon embeddings were evaluated with linear and nonlinear regression models to determine how much information about compound-induced pathway responses could be recovered from molecular structure alone (Burns et al. 2025).

A major challenge to structure-based prediction is that structural similarity does not always imply similar biological activity. Small chemical modifications can sometimes produce large changes in potency or function, a phenomenon commonly described as an activity cliff (Tilborg et al. 2022). An analogous pattern may occur at the pathway level if structurally similar compounds produce substantially different inferred pathway responses. Characterizing these pathway-response differences may help identify regions of chemical space in which structure-based models are unreliable and may reveal structural modifications associated with changes in cellular signaling.

Large-scale perturbational transcriptomic data present several challenges for structure-to-response modeling. Measurements obtained from different experimental blocks may contain systematic technical variation, and repeated measurements of the same compound may remain inconsistent after block adjustment. This within-compound variability limits the reproducible biological signal available for prediction. Moreover, molecular structure alone does not describe target engagement, cellular exposure, metabolism, off-target interactions, or other context-dependent factors that can influence transcriptional responses. These issues are considered when interpreting model performance and are examined further in the Limitations section.

The primary objective of this study was to determine whether molecular structure can predict compound-induced pathway responses in A549 cells for previously unobserved structures. A secondary objective was to investigate whether structurally similar compounds can produce substantially different pathway-response profiles. The analysis is limited to transcriptionally inferred pathway responses under one cell line, dose, and exposure duration. Because outcomes such as cell viability, growth inhibition, and half-maximal inhibitory concentration (IC_{50}) were not included, the study does not establish whether the predicted pathway changes correspond to anticancer efficacy. Connecting pathway responses with therapeutic outcomes remains an important direction for future research.

Methods

Data Description

Gene-expression data were obtained from the 2020 beta release of the Library of Integrated Network-based Cellular Signatures (LINCS) Connectivity Map L1000 resource (Subramanian et al. 2017). We used the Level 5 compound-perturbation matrix together with the accompanying signature and gene metadata. In the L1000 processing pipeline, a Level 5 signature is a replicate-consensus differential-expression profile produced by moderated z-score (MODZ) aggregation of replicate-level profiles for the same perturbation condition (Subramanian et al. 2017). Replicates that are more concordant with the other replicates receive greater weight. Thus, each Level 5 signature represents an aggregated response to a compound under a particular experimental condition.

The analysis was restricted to small-molecule compound perturbations in A549 human lung adenocarcinoma cells. We retained signatures that passed the LINCS quality-control criteria and were designated high quality. To provide a consistent cellular context, signatures were further restricted to an exposure duration of 24 h and the nominal 10 μ M dose category. This metadata selection produced 2,418 Level 5 signatures. After requiring a usable molecular structure and completing the structure-curation procedure described below, the analytical dataset contained 2,355 signatures representing 1,498 LINCS compound identifiers and 1,494 distinct standardized molecular structures.

The source expression matrix contained 12,328 L1000 gene features, comprising directly measured landmark genes and computationally inferred genes. These high-dimensional signatures were converted into pathway-level responses using PROGENy, which estimates pathway activity from downstream pathway-responsive genes. (Schubert et al. 2018; Holland et al. 2020). Using the human PROGENy model, the 100 highest-ranked responsive genes for each pathway were selected, and a signed pathway score was calculated as

$$A_{sp} = \sum_{g \in G_p} w_{gp} E_{sg},$$

where E_{sg} is the Level 5 expression value for gene g in signature s , w_{gp} is the PROGENy weight connecting gene g to pathway p , and G_p is the selected response-gene set for pathway p . Weighted sums were calculated using the decoupler implementation (Badia-i-Mompel et al. 2022). This procedure produced one 14-dimensional response vector for each signature, comprising Androgen, EGFR, Estrogen, Hypoxia, JAK-STAT, MAPK, NF- κ B, PI3K, TGF- β , TNF- α , TRAIL, VEGF, WNT, and p53 pathway scores. Positive and negative values indicate expression patterns consistent with relatively increased or decreased pathway activity, respectively; they are transcriptionally inferred scores rather than direct biochemical measurements.

Molecular-Structure Collection and Standardization

The Simplified Molecular Input Line Entry System (SMILES) encodes a molecular structure as a linear text string. Atomic symbols represent the atoms in the molecular graph, while additional notation specifies bonds, branching, ring closures, formal charges, and, when known, tetrahedral or double-bond stereochemistry; hydrogen atoms are generally represented implicitly. A SMILES string is therefore a machine-readable description of molecular structure. In the downstream analysis, the curated SMILES were converted into two numerical molecular representations: Morgan fingerprints and pretrained CheMeleon embeddings.

Chemical structures were linked to the filtered LINCS signatures using the LINCS perturbagen identifier (`pert_id`). SMILES annotations were first obtained from the 2020 LINCS release. Among the 1,555 compound identifiers satisfying the expression-data criteria, 107 had a missing entry in the current LINCS compound metadata. For compounds without a current structure annotation, earlier LINCS perturbagen records from GEO accessions GSE92742 and GSE70138 were searched using the same identifier, recovering two additional structures (Subramanian et al. 2017).

The remaining missing structures were queried against PubChem (Kim et al. 2025). A PubChem candidate was accepted only when its synonym list contained an exact match to the core BRD identifier. This procedure recovered structures for 55 additional compounds; 50 compounds remained without a recoverable structure and were excluded.

Structural standardization was necessary and applied because a molecule does not have a unique raw SMILES representation. Equivalent molecular graphs can be written using different atom orderings, while database records may also differ in their inclusion of salts, solvents, protonation states, functional-group conventions, or stereochemical annotations. Without consistent processing, the same underlying compound could consequently be assigned different numerical representations, or chemically distinct records could be treated as equivalent. The objective of the curation procedure was therefore to produce a consistent, structure-preserving SMILES representation for each compound.

All non-missing structure entries were processed using a custom curation pipeline implemented with RDKit and `chembl_structure_pipeline` (RDKit contributors 2026; Bento et al. 2020). Raw stereochemical annotations were audited before structural cleanup because parsing and sanitization can remove ineffective or malformed atom- and bond-level annotations. Each SMILES was then parsed and sanitized, atom-map identifiers were removed, and functional-group representations were normalized. Parent structures were generated using the ChEMBL list-based salt and solvent definitions (Bento et al. 2020). Fragment selection was based on membership in these curated lists, rather than solely on fragment size. Records that remained multicomponent after parent selection were withheld for manual review.

Eligible parent structures were neutralized through proton addition or removal when possible, while permanent charges were preserved. This operation standardized neutralizable charge states but was not intended to predict the predominant protonation state under physiological

conditions. The resulting structures were written as canonical isomeric SMILES and required to pass an RDKit parse-and-canonicalization round-trip.

Automated screening placed 68 compound identifiers, corresponding to 97 signature rows, into review. Review flags included incomplete directional-bond annotations that did not define valid double-bond stereochemistry, atom-level chirality annotations lost during parsing, unresolved multicomponent structures, unusual chemistry, and identical standardized structures associated with different LINCX identifiers. Cleaned candidates were retained for 13 compounds whose ineffective atom-stereo annotations were removed during parsing. Distinct LINCX identifiers that resolved to the same standardized structure were also retained because they provided repeated biological observations of the same model input; they were assigned a common structure identifier during subsequent modeling.

Compounds with incomplete or unusable directional annotations were compared with independently retrieved PubChem records. Candidate structures were first compared after removing stereochemical annotations to determine whether their underlying atom connectivity, formal charges, and bond orders agreed. When the non-stereochemical molecular graphs matched, compatibility with the stereochemistry already specified by LINCX was evaluated before accepting additional stereochemical information from PubChem. This procedure resolved 40 compounds. Five compounds for which the LINCX and PubChem records remained missing or structurally inconsistent were excluded, as were two entries represented only by restricted structure placeholders.

A second stereochemical audit identified potential tetrahedral centers and double bonds whose configurations remained unspecified. Resolving these annotations was important because different stereoisomers can differ substantially in target binding and biological activity. Unspecified stereochemistry can therefore make the molecular input ambiguous and obscure structure-pathway relationships. Nevertheless, unspecified stereochemistry was not considered an error by itself because a database record may intentionally represent a racemate, a stereoisomeric mixture, or an unassigned configuration.

PubChem was searched for compounds with multiple unresolved stereocenters, partially specified stereochemistry, or unresolved double-bond geometry. A PubChem structure was accepted as a stereochemical refinement only when it had the same canonical molecular graph after removal of stereochemistry, preserved all compatible stereochemistry already present in the curated LINCX structure, and reduced the number of unspecified stereochemical elements. Candidates exhibiting a graph or tautomer difference, a stereochemical conflict, or no improvement in stereochemical completeness were not used as replacements.

This refinement replaced the structures of 107 compounds with more complete, nonconflicting PubChem records. Of these, 91 became fully specified with respect to all stereochemical elements detected by RDKit, while 16 retained some unresolved stereochemistry but were more complete than their original records. The final dataset contained 1,498 compound identifiers mapped to 1,494 distinct standardized structures. Among these compounds, 1,205 had no

detected unresolved stereochemical elements, whereas 293 retained at least one unresolved element. The latter were retained with explicit audit flags so that residual structural uncertainty could be considered when interpreting model performance.

The resulting canonical isomeric SMILES served as the common structural input for the downstream representation stage. Each distinct standardized structure was converted into a chirality-aware Morgan fingerprint and a pretrained CheMeleon molecular embedding, as described in the following subsection. This ensured that differences between the two numerical representations reflected their respective encoding methods rather than avoidable inconsistencies in the underlying chemical records.

Molecular Representations

Each of the 1,494 distinct standardized molecular structures was converted into two fixed-length numerical representations: a Morgan fingerprint and a pretrained CheMeleon embedding. Representations were generated once for each unique standardized structure and reused for all corresponding LINCX signatures. .

Morgan fingerprints provide a conventional, rule-based representation of molecular structure (Rogers and Hahn 2010). Using RDKit, we generated a 2,048-bit binary fingerprint with a radius of two bonds and stereochemical information enabled (RDKit contributors 2026). The algorithm identifies circular atom environments surrounding each atom and hashes these environments into a fixed-length bit vector, in which an active bit indicates the presence of one or more corresponding substructures. Morgan fingerprints provide a transparent and computationally efficient representation of local chemical patterns, but they may lose information through hash collisions and do not explicitly model global molecular context or three-dimensional conformation.

CheMeleon provides a learned graph-based representation derived from a pretrained directed message-passing neural network (Burns et al. 2026). The CheMeleon encoder was pretrained on approximately one million PubChem molecules to predict deterministic Mordred molecular descriptors, allowing it to learn transferable structural features before application to the present dataset. For each standardized SMILES, molecular-graph information was propagated along directed bonds, and the resulting atom-level features were combined by mean aggregation to produce a 2,048-dimensional continuous embedding. The published encoder checkpoint was used with frozen weights: no parameters were updated using the A549 pathway-response data. CheMeleon therefore served only as a molecular feature extractor rather than as the final pathway-prediction model.

The two representations offer complementary descriptions of the same standardized structures. Morgan fingerprints encode predefined local substructures as sparse binary features, whereas CheMeleon embeddings encode learned structural patterns as dense continuous features. Comparing models constructed from these representations allowed us to assess whether pretrained

graph-based features captured structure–pathway relationships that were not adequately represented by a conventional circular fingerprint.

Both representations provided one predictor vector for each standardized structure. The biological response data, however, remained at the signature level, and some structures were associated with multiple signatures collected in different experimental blocks. Before fitting the predictive models, these repeated pathway measurements were therefore adjusted for systematic block variation and aggregated to obtain a single pathway-response vector for each standardized structure.

Adjustment for Experimental Block Effects and Structure-Level Response Estimation

Before block effects were tested or estimated, the 1,494 standardized structures were partitioned into training, validation, and test sets. All signatures and LINCS compound identifiers associated with the same standardized structure were assigned to the same partition. This structure-level partitioning ensured that the validation and test sets contained molecular structures not used during model development and prevented repeated observations of the same structure from appearing in different partitions.

A 70:15:15 allocation was generated. The resulting training, validation, and test sets contained 1,045, 224, and 225 structures, respectively, corresponding to 1,587, 446, and 322 Level 5 signatures. All outcome-dependent preprocessing—including block-effect testing, selection of shrinkage penalties, and estimation of block coefficients—was performed using the training data alone, since using the complete dataset for these procedures would allow validation and test pathway scores to influence the construction of their own response variables, potentially biasing estimates of predictive performance.

Although cell line, dose category, and exposure duration were held constant, multiple Level 5 signatures could correspond to the same standardized molecular structure. These signatures were sometimes generated in different experimental blocks, which were operationally defined by the prefix of `sig_id` preceding the first colon. This prefix corresponds to the LINCS brew prefix or replicate set. The analytical dataset contained 68 such blocks. Systematic technical differences among blocks could propagate from the gene-expression measurements into the inferred pathway scores and consequently be mistaken for structure-associated biological responses. Moreover, treating every signature as an independent observation would give disproportionate weight to structures or blocks represented by more signatures.

Signatures were therefore first averaged within each structure–block combination. For pathway p , the mean response of structure d in block b was defined as

$$\bar{Y}_{dbp} = \frac{1}{n_{db}} \sum_{j=1}^{n_{db}} Y_{dbjp},$$

where Y_{dbjp} is the PROGENy score for signature j , and n_{db} is the number of signatures associated with that structure–block combination. This produced 2,208 structure–block pairs across the complete dataset, including 1,576 pairs in the training partition.

A block effect can be separated from a structure effect only when at least one structure has been observed in multiple blocks. Accordingly, block-effect estimation was restricted to the 221 training structures represented in at least two distinct blocks. These anchor structures contributed 752 structure–block pairs. A bipartite graph connecting structures and experimental blocks was constructed to determine which block effects were mutually identifiable through shared structures. The largest connected component contained all 221 anchor structures and 63 of the 68 blocks and was used as the reference network. Blocks disconnected from this network could not be compared with the reference blocks and were therefore assigned a zero adjustment. Connected blocks represented by fewer than three anchor structures were retained under shrinkage but flagged as weakly supported.

For each pathway, evidence of systematic block-associated variation was evaluated by comparing a reduced model containing structure fixed effects with a full model containing both structure and block fixed effects:

$$\begin{aligned} \text{Reduced:} \quad & \bar{Y}_{dbp} = \alpha_{dp} + \varepsilon_{dbp}, \\ \text{Full:} \quad & \bar{Y}_{dbp} = \alpha_{dp} + \beta_{bp} + \varepsilon_{dbp}, \end{aligned}$$

where α_{dp} represents the mean response associated with structure d , and β_{bp} represents a shared shift associated with block b . Structure effects were removed by within-structure centering before testing. A classical partial F -test was calculated, together with a permutation test in which complete 14-pathway response vectors were permuted among blocks within each repeated training structure. This permutation scheme preserved the correlations among pathway scores while breaking their association with experimental block. A total of 9,999 permutations were performed, and the resulting pathway-level p -values were adjusted for multiple testing using the Benjamini–Hochberg procedure (Benjamini and Hochberg 1995).

The p - and q -values were used only to characterize the evidence for block-associated variation. They were not used to determine whether a pathway was adjusted or to select the magnitude of shrinkage. Instead, block effects were estimated for every pathway, with the degree of regularization selected independently by cross-validation.

Block effects were estimated using a ridge-regularized additive model:

$$\underset{\alpha, \beta}{\text{minimize}} \left\{ \sum_{d,b} (\bar{Y}_{dbp} - \alpha_{dp} - \beta_{bp})^2 + \lambda_p \sum_b \beta_{bp}^2 \right\}.$$

The structure effects α_{dp} were not penalized, whereas the block effects β_{bp} were shrunk toward zero (Hoerl and Kennard 1970). This regularization reduced the instability of estimates for

blocks supported by relatively few shared structures. A separate penalty λ_p was selected for each pathway by five-fold cross-validation among the repeated structures. Structure–block observations were held out during cross-validation, and each structure was assigned a total validation weight of one so that highly repeated structures did not dominate penalty selection. After selection, each pathway-specific model was refitted using all 752 connected training structure–block pairs.

The estimated block effects were then frozen and applied without re-estimation to the training, validation, and test observations. Observations from blocks that were absent from or disconnected within the training reference network received a zero adjustment. Finally, the adjusted structure–block scores were averaged with equal weight across the blocks available for each structure:

$$\frac{1}{B_d} \sum_{b \in \mathcal{B}_d} (\bar{Y}_{dbp} - \hat{\beta}_{bp}),$$

where $\mathcal{B}_d$ is the set of blocks in which structure d was observed and $B_d = |\mathcal{B}_d|$. Equal block weighting prevented blocks containing more signatures from exerting greater influence on the final response. This procedure produced one adjusted 14-pathway response vector for each of the 1,494 standardized structures, thereby aligning the response matrix with the structure-level Morgan and CheMeleon predictor matrices.

Yes—the next subsection should describe the predictive models, preprocessing, and model-selection procedure. Numerical performance belongs in the Results section.

Predictive Models and Model Selection

The structure-level molecular representations and adjusted pathway-response vectors were used to learn a mapping from molecular structure to pathway activity. For structure d , let $\mathbf{x}_d \in \mathbb{R}^{2048}$ denote either its Morgan fingerprint or CheMeleon embedding, and let $\mathbf{y}_d \in \mathbb{R}^{14}$ denote its adjusted PROGENy response vector. Pathway responses were standardized using the training-set mean and standard deviation:

$$\tilde{y}_{dp} = \frac{y_{dp} - \mu_{p,\text{train}}}{\sigma_{p,\text{train}}},$$

where p indexes pathways. CheMeleon features were also standardized using training-set statistics, whereas the binary Morgan fingerprints were retained on their original scale.

Four regression approaches were evaluated. Ridge regression provided a regularized linear reference model for both molecular representations (Hoerl and Kennard 1970). Its coefficients were estimated by

$$(\hat{\mathbf{B}}, \hat{\mathbf{b}}) = \underset{\mathbf{B}, \mathbf{b}}{\operatorname{argmin}} \left\{ \left\| \tilde{\mathbf{Y}} - \mathbf{1}\mathbf{b}^\top - \mathbf{X}\mathbf{B} \right\|_F^2 + \alpha \|\mathbf{B}\|_F^2 \right\},$$

where $\mathbf{B}$ contains the feature coefficients for the 14 pathways and α controls coefficient shrinkage.

Kernel ridge regression was used to model nonlinear structure–response relationships. For each pathway, the dual coefficients and prediction function were

$$\hat{\mathbf{c}}_p = (\mathbf{K} + \alpha_p \mathbf{I})^{-1} \tilde{\mathbf{y}}_p, \quad \hat{f}_p(\mathbf{x}) = \mathbf{k}(\mathbf{x}, \mathbf{X})^\top \hat{\mathbf{c}}_p.$$

CheMeleon embeddings were compared using the radial basis function kernel

$$k_{\text{RBF}}(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma_p \|\mathbf{x}_i - \mathbf{x}_j\|_2^2),$$

whereas Morgan fingerprints were compared using the Tanimoto kernel

$$k_{\text{Tan}}(\mathbf{x}_i, \mathbf{x}_j) = \frac{\mathbf{x}_i^\top \mathbf{x}_j}{\|\mathbf{x}_i\|_1 + \|\mathbf{x}_j\|_1 - \mathbf{x}_i^\top \mathbf{x}_j}.$$

The RBF kernel measures proximity between continuous CheMeleon embeddings, while the Tanimoto kernel measures the proportion of shared fingerprint features.

Pathway-specific support-vector regression models were also fitted to the CheMeleon embeddings using an RBF kernel (Drucker et al. 1997). For pathway p , the model minimized

$$\frac{1}{2} \|f_p\|_{\mathcal{H}}^2 + C_p \sum_d L_{\epsilon_p}(\tilde{y}_{dp} - f_p(\tilde{\mathbf{x}}_d)),$$

where

$$L_{\epsilon_p}(r) = \max(0, |r| - \epsilon_p).$$

The parameter C_p controls the penalty assigned to prediction errors, while ϵ_p defines an error-insensitive region around the fitted function.

Finally, a feed-forward neural network was trained on the CheMeleon embeddings to predict all 14 pathways jointly. The selected architecture contained one hidden layer with layer normalization, GELU activation, and dropout:

$$\mathbf{h}_d = \text{Dropout}[\text{GELU}(\text{LayerNorm}(\mathbf{W}_1 \tilde{\mathbf{x}}_d + \mathbf{b}_1))], \quad \hat{\tilde{\mathbf{y}}}_d = \mathbf{W}_2 \mathbf{h}_d + \mathbf{b}_2.$$

The network was optimized using mean squared error and AdamW, with early stopping based on validation performance. Predictions from five independently initialized networks were averaged to reduce sensitivity to random initialization (Paszke et al. 2019).

Predictive performance was based on standardized root mean squared error:

$$\text{sRMSE} = \sqrt{\frac{1}{14n} \sum_{d=1}^n \sum_{p=1}^{14} \left(\frac{y_{dp} - \hat{y}_{dp}}{\sigma_{p,\text{train}}} \right)^2}.$$

Predictive performance alone does not characterize the properties of pathway responses that may influence their predictability from molecular structure. Two relevant properties are the reproducibility of responses across experimental blocks for the same structure and the occurrence of local discontinuities, where structurally similar compounds produce substantially different responses. To complement predictive evaluation, we conducted two diagnostic analyses. Conditional cross-block reliability assessed whether differences in pathway response between structures were consistent across blocks, after accounting for shared block effects. An activity-cliff analysis examined whether highly similar structures exhibited unusually divergent pathway-response profiles.

Conditional Cross-Block Pathway Reliability

Conditional cross-block reliability measures the extent to which differences in pathway response between structures persist across experimental blocks after accounting for shared block effects. When measurements of the same structure vary substantially relative to differences between structures, the response provides a less consistent target for learning. Such variability can obscure structure–response relationships during training and increase prediction error during testing, because a structure-only model cannot predict fluctuations unrelated to molecular structure. Conversely, high reliability indicates stable differences between structures, although it does not guarantee that a particular molecular representation or model can capture those differences. Therefore, we conducted a conditional cross-block reliability analysis to quantify response reproducibility and provide context for interpreting predictive performance.

Reliability was estimated from the unadjusted signature-level PROGENy scores. Signatures associated with the same structure and experimental block were first averaged:

$$\bar{Y}_{dbp} = \frac{1}{n_{db}} \sum_{r=1}^{n_{db}} Y_{dbrp},$$

where Y_{dbrp} is the pathway score for replicate signature r of structure d in block b , and n_{db} is the number of corresponding signatures. The primary analysis was restricted to structures observed in at least two blocks within the largest connected component of the structure–block network.

For each pathway p , the structure–block mean scores were modeled using a linear mixed-effects model:

$$\bar{Y}_{dbp} = \mu_p + \beta_{bp} + a_{dp} + \varepsilon_{dbp},$$

where $\bar{Y}_{dbp}$ is the mean response of structure d in block b , μ_p is the overall pathway mean, and β_{bp} is a fixed effect representing the shared shift associated with block b . The random structure effect and residual error were assumed to follow

$$a_{dp} \sim N(0, \tau_p^2), \quad \varepsilon_{dbp} \sim N(0, \sigma_p^2),$$

with

$$\text{Cov}(a_{dp}, \varepsilon_{dbp}) = 0.$$

Here, τ_p^2 represents stable variation between structures, whereas σ_p^2 represents the remaining within-structure variation across blocks after accounting for shared block effects. The residual component may include experimental variability, structure-by-block interactions, and other unmeasured factors and therefore should not be interpreted as purely technical error.

Variance components were estimated separately for each pathway using restricted maximum likelihood. Conditional reliability for a single structure–block observation was summarized using the intraclass correlation coefficient:

$$ICC_p = \frac{\tau_p^2}{\tau_p^2 + \sigma_p^2}.$$

Values approaching one indicate that most of the variation remaining after adjustment for shared block effects is attributable to stable differences between structures. Values approaching zero indicate that within-structure variation across blocks is large relative to between-structure variation. Approximate 95% confidence intervals were obtained using 1,000 parametric-bootstrap repetitions, in which pathway responses were simulated from the fitted model while preserving the observed structure–block design, after which the model was refitted to each simulated dataset.

The reliability model assumes additive block effects shared across structures, normally distributed and exchangeable structure effects, and independent residuals with constant variance. It also assumes that the repeatedly measured structures are informative about the broader compound collection.

Pathway-Response Activity-Cliff Analysis

A complementary analysis examined whether structurally similar compounds produced unexpectedly different pathway response. Shared block effects were re-estimated using the pooled dataset. The block-adjusted response was defined as:

$$\widetilde{Y}_{dbp} = \overline{Y}_{dbp} - \hat{\beta}_{bp},$$

where $\hat{\beta}_{bp}$ is the estimated fixed effect of block b for pathway p . The adjusted structure–block responses were then averaged with equal weight across the blocks in which each structure was observed:

$$A_{dp} = \frac{1}{|\mathcal{B}_d|} \sum_{b \in \mathcal{B}_d} \widetilde{Y}_{dbp},$$

where $\mathcal{B}_d$ is the set of experimental blocks containing structure d . This procedure produced one 14-pathway response vector for each standardized structure.

Each pathway was standardized across all N structures.

$$Z_{dp} = \frac{A_{dp} - \overline{A}_p}{s_p},$$

where $\overline{A}_p, s_p$ are pathway-specific mean and standard deviation, respectively.

Structural similarity between structures i and j was calculated from radius-2, 2,048-bit Morgan fingerprints using the Tanimoto coefficient (Rogers and Hahn 2010; RDKit contributors 2026):

$$S_{ij} = \frac{\mathbf{x}_i^T \mathbf{x}_j}{\|\mathbf{x}_i\|_1 + \|\mathbf{x}_j\|_1 - \mathbf{x}_i^T \mathbf{x}_j},$$

where $\mathbf{x}_i$ and $\mathbf{x}_j$ are the binary Morgan-fingerprint vectors of structures i and j . The set of structurally similar pairs eligible for cliff assessment was defined as

$$\mathcal{E} = \{(i, j) : i < j, S_{ij} \geq 0.80\}.$$

For each unique pair, the overall pathway-response distance was defined as the root mean squared difference across the $P = 14$ standardized pathway scores:

$$D_{ij} = \sqrt{\frac{1}{P} \sum_{p=1}^P (Z_{ip} - Z_{jp})^2}, \quad P = 14.$$

Thus, D_{ij} summarizes the overall divergence between the pathway-response profiles of structures i and j , with larger values indicating more dissimilar standardized pathway responses.

The global response-divergence threshold was defined as

$$q_{0.95}^{(D)} = \text{Quantile}_{0.95}(\{D_{ij} : i < j\}).$$

A global candidate pathway-response cliff was identified using the indicator

$$C_{ij}^{(\text{global})} = \mathbb{I}(S_{ij} \geq 0.80 \text{ and } D_{ij} > q_{0.95}^{(D)}),$$

where $\mathbb{I}(\cdot)$ equals one when the stated conditions are satisfied and zero otherwise.

For each pathway p , the pairwise absolute standardized response difference was defined as

$$\Delta_{ijp} = |Z_{ip} - Z_{jp}|,$$

and its pathway-specific threshold was defined as

$$q_{0.95,p}^{(\Delta)} = \text{Quantile}_{0.95}(\{\Delta_{ijp} : i < j\}).$$

A pathway-specific candidate cliff was then identified using

$$C_{ijp}^{(\text{pathway})} = \mathbb{I}(S_{ij} \geq 0.80 \text{ and } \Delta_{ijp} > q_{0.95,p}^{(\Delta)}).$$

A fixed two-standard-deviation sensitivity criterion was also examined:

$$C_{ijp}^{(2\text{SD})} = \mathbb{I}(S_{ij} \geq 0.80 \text{ and } \Delta_{ijp} \geq 2).$$

Because approximately 5% of all unique pairs exceed each empirical 95th-percentile threshold by construction, interpretation focused on whether extreme pathway-response differences were more frequent among structurally similar pairs than among all structure pairs.

Results

Experimental Block Variation

Before evaluating structure-based predictions, we examined whether pathway responses varied systematically across experimental blocks after accounting for molecular structure identity. The training reference network contained 752 structure–block observations from 221 structures across 63 blocks. The pathway-specific results are presented in Table 1.

Table 1: Experimental-block variation conditional on structure identity in the training data. Partial R^2 denotes the proportion of residual variation explained by adding block effects to a model containing structure fixed effects. Permutation tests used 9,999 within-structure permutations, with Benjamini–Hochberg adjustment across 14 pathways.

Pathway	Partial R^2	Permutation p	BH-adjusted q
MAPK	0.306	1.00e-04	3.50e-04
Hypoxia	0.303	1.00e-04	3.50e-04
PI3K	0.266	1.00e-04	3.50e-04
p53	0.232	1.00e-04	3.50e-04
EGFR	0.233	2.00e-04	5.60e-04
TRAIL	0.182	3.00e-04	7.00e-04
Androgen	0.166	0.0120	0.0240
TNF- α	0.160	0.0169	0.0296
TGF- β	0.158	0.0263	0.0409
JAK–STAT	0.153	0.0343	0.0480
VEGF	0.150	0.0401	0.0510
NF- κ B	0.150	0.0562	0.0656
Estrogen	0.143	0.1427	0.1537
WNT	0.115	0.5340	0.5340

Block-associated variation was detected in 10 of the 14 pathways at a Benjamini–Hochberg-adjusted threshold of $q < 0.05$. The largest conditional effects occurred in MAPK, Hypoxia, and PI3K, for which block terms explained 30.6%, 30.3%, and 26.6%, respectively, of the variation remaining after accounting for structure identity. All three pathways had permutation $p = 0.0001$ and adjusted $q = 0.00035$. These partial R^2 values describe the fit of the diagnostic block models; they do not measure predictive performance or the amount of variation removed by the subsequent regularized adjustment.

VEGF, NF- κ B, Estrogen, and WNT did not meet the adjusted significance threshold, with q values of 0.051, 0.066, 0.154, and 0.534, respectively. These results provide weaker evidence of shared block effects for these pathways.

The findings indicate that shared block-associated differences persisted despite restricting the analysis to a common cell line, dose category, and exposure duration. This motivates the removal of block effects using the independently selected shrinkage penalties described in Methods, when constructing structure-level responses. However, the significance tests themselves were not used to select pathways for adjustment.

Predictive Performance

After accounting for shared experimental-block variation and constructing structure-level responses, we evaluated how accurately molecular representations predicted pathway scores for previously unseen structures. Table 2 summarizes the performance of the linear and nonlinear models on the 225 test structures.

Table 2: Predictive performance on test set’s molecular structures. Bold indicates the model with the highest mean test R^2 . Mean test R^2 is the average across the 14 pathways. Higher R^2 and lower sRMSE indicate better performance. KRR: kernel ridge regression; SVR: support-vector regression; FNN: feedforward neural network.

Representation	Model	Mean test R^2	Test sRMSE
Morgan	Ridge	0.079	0.983
CheMeleon	Ridge	0.052	1.002
Morgan	Tanimoto KRR	0.102	0.969
CheMeleon	RBF KRR	0.150	0.943
CheMeleon	RBF SVR	0.131	0.952
CheMeleon	FNN ensemble	0.115	0.967

Overall predictive performance was limited, with mean test R^2 values ranging from 0.052 to 0.150. The nonlinear models achieved higher mean test R^2 values than either linear ridge model. CheMeleon with RBF kernel ridge regression achieved the highest observed mean test R^2 of 0.150 and the lowest sRMSE of 0.943, followed by CheMeleon with RBF support-vector regression ($R^2 = 0.131$). The FNN ensemble and Morgan Tanimoto kernel ridge model achieved mean test R^2 values of 0.115 and 0.102, respectively.

The value of the molecular representation depended on the accompanying regression method. With linear ridge regression, Morgan fingerprints outperformed CheMeleon embeddings (mean test $R^2 = 0.079$ versus 0.052). In contrast, with nonlinear kernel regression, CheMeleon RBF KRR outperformed Morgan Tanimoto KRR (0.150 versus 0.102). CheMeleon therefore did not provide a uniformly better representation; its strongest performance emerged only when paired with a nonlinear, pathway-specific model.

Table 3: Pathway-specific test performance of the best overall model, CheMeleon with pathway-specific RBF kernel ridge regression. Higher R^2 and lower RMSE or sRMSE indicate better performance.

Pathway	Test RMSE	Test sRMSE	Test R^2
MAPK	126.569	0.982	0.361
EGFR	80.001	1.063	0.303
p53	188.225	0.906	0.222
PI3K	92.949	0.758	0.215
TGF- β	76.309	0.841	0.179
JAK-STAT	260.880	0.923	0.175
NF- κ B	335.544	1.084	0.111
Hypoxia	232.489	0.844	0.108
Estrogen	74.413	0.860	0.106
TNF- α	339.742	1.119	0.099
TRAIL	15.267	0.902	0.064
VEGF	36.814	0.944	0.060
WNT	27.835	0.908	0.054
Androgen	91.183	0.989	0.042

The pathway-specific results also show that the overall mean masks substantial heterogeneity. MAPK had the highest test R^2 (0.361; sRMSE = 0.982), whereas Androgen had the lowest test R^2 (0.042; sRMSE = 0.989). Nevertheless, even the best model left most pathway variation unexplained. This limitation could reflect irreproducible pathway measurements, structural representations that omit biologically relevant information, or genuinely discontinuous structure-response relationships. We therefore next examined reproducibility of pathway responses across experimental blocks and the occurrence of unusually divergent responses among structurally similar compounds.

Conditional Cross-Block Pathway Reliability

As described in Methods, conditional cross-block reliability assesses whether differences in pathway response between structures persist across experimental blocks after accounting for shared block effects. To explain the limited predictive performance reported above, this analysis helps assess whether inconsistent responses to the same structure could contribute to prediction error.

The reliability analysis used full data, including 316 structures contributing 1030 structure-block means across 63 experimental blocks. Table 4 reports the conditional ICC estimates and approximate 95% confidence intervals, alongside pathway-specific test R^2 for the CheMeleon RBF kernel ridge model, which had the highest observed mean test R^2 in the preceding comparison. Figure 2 explores the association between ICC and R^2 .

Table 4: Conditional cross-block pathway reliability and descriptive comparison with CheMeleon RBF kernel ridge test performance. Intervals are approximate 95% confidence intervals obtained from 1,000 parametric-bootstrap repetitions.

Pathway	Conditional ICC	95% CI	Test R^2
MAPK	0.847	[0.816, 0.870]	0.361
p53	0.844	[0.815, 0.868]	0.222
EGFR	0.815	[0.781, 0.842]	0.303
PI3K	0.764	[0.724, 0.800]	0.215
Hypoxia	0.751	[0.709, 0.787]	0.108
TNF- α	0.743	[0.695, 0.783]	0.099
NF- κ B	0.722	[0.673, 0.764]	0.111
TGF- β	0.629	[0.572, 0.679]	0.179
JAK-STAT	0.605	[0.546, 0.655]	0.175
WNT	0.548	[0.479, 0.603]	0.054
TRAIL	0.533	[0.463, 0.592]	0.064
Estrogen	0.533	[0.464, 0.590]	0.106
VEGF	0.514	[0.443, 0.580]	0.060
Androgen	0.413	[0.343, 0.482]	0.042

Reliability varied substantially across pathways. Conditional ICC ranged from 0.413 for Androgen (95% CI: 0.343–0.482) to 0.847 for MAPK (0.816–0.870), with similarly high estimates for p53 (0.844; 0.815–0.868) and EGFR (0.815; 0.781–0.842). Intervals were obtained from 1,000 parametric-bootstrap repetitions. .

Figure 2 showed a positive association between conditional ICC and pathway-specific test R^2 for CheMeleon RBF kernel ridge regression (Spearman $\rho = 0.86$ across 14 pathways). More reproducible pathways generally had higher predictive performance. MAPK, for example, combined the highest estimated ICC with a test R^2 of 0.361, whereas Androgen had both lowest reliability and predictive performance. This pattern is consistent with response reproducibility contributing to differences in predictability across pathways, but does not establish a causal relationship.

ICC and R^2 nevertheless characterize different aspects of the response. ICC measures whether differences between structures persist across experimental blocks, without using molecular fingerprints; test R^2 measures how accurately those responses can be predicted for unseen structures from their molecular representations. High reliability therefore indicates a stable response component, but does not guarantee that the available features and models can capture it.

Together, these findings suggest that response inconsistency may contribute to limited prediction, while leaving room for additional limitations of the molecular representations, models, and available training data.

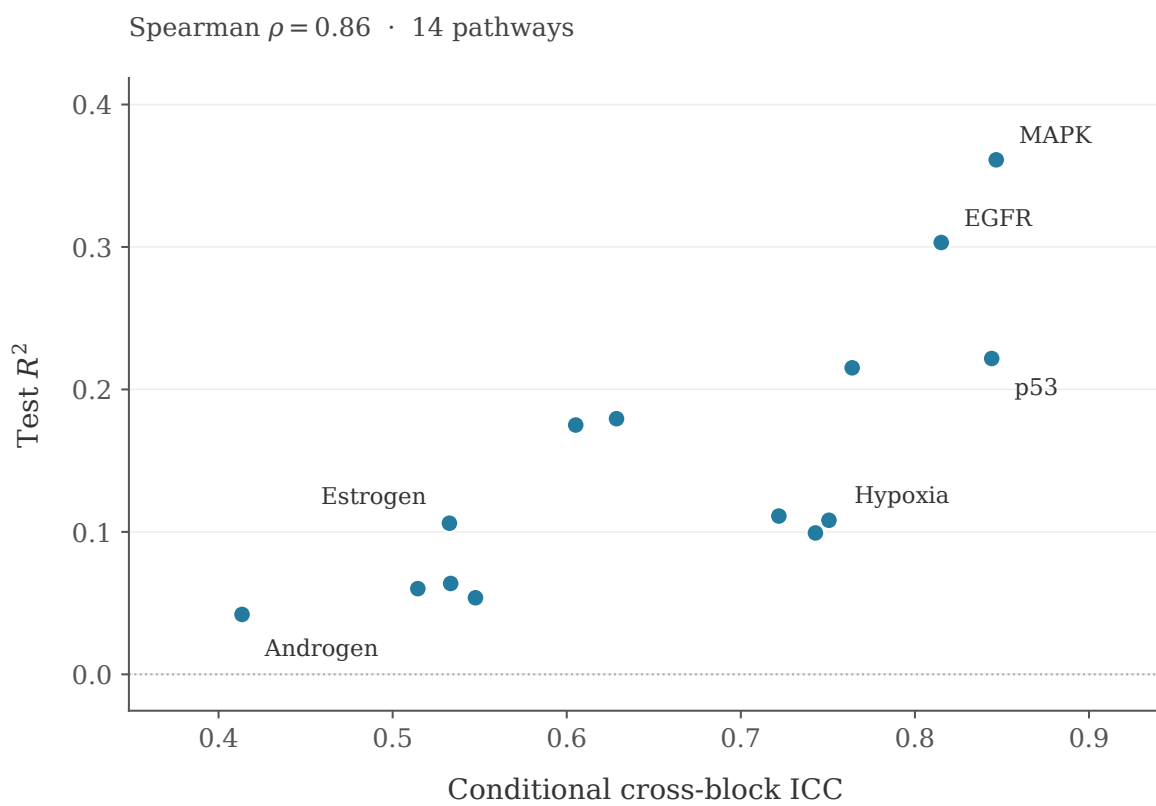


Figure 2: Descriptive association between conditional cross-block ICC and pathway-specific test R^2 for CheMeleon RBF kernel ridge regression.

Pathway-Response Activity Cliffs

Beyond response reproducibility, local differences between structurally similar compounds offer another potential source of prediction difficulty. If similar molecular structures produce markedly different pathway responses, models that learn smooth structure-response relationships may struggle to distinguish them. We therefore examined candidate pathway-response activity cliffs using the full dataset.

As shown in Figure 3, among 1,494 structures, 237 pairs involving 201 structures met the Morgan Tanimoto similarity threshold of 0.80. None exceeded the global response-distance threshold of 2.592, defined by the 95th percentile of distances across all unique structure pairs. However, 23 eligible pairs (9.7%), involving 32 distinct structures, exceeded a pathway-specific threshold for at least one pathway. Thus, marked differences in individual pathways were present even though no pair met the criterion for unusually divergent overall response profiles.

TNF- α and NF- κ B had the most candidate cliff pairs (8 each; 3.4% of eligible pairs), followed by Estrogen and WNT (5 each; 2.1%) and TGF- β (4; 1.7%). No candidate cliffs were identified for MAPK or EGFR.

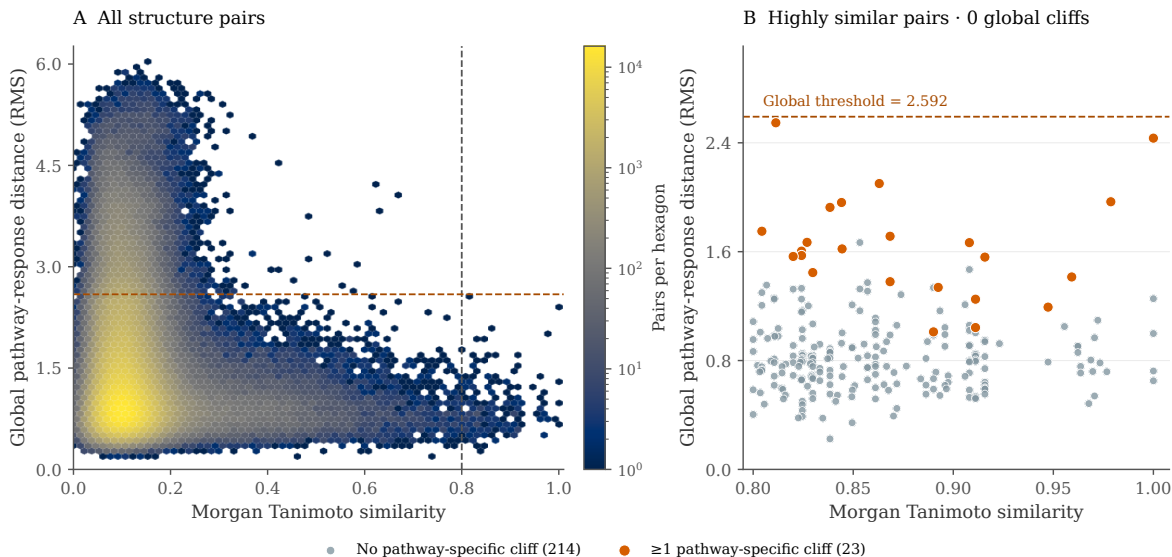


Figure 3: Molecular similarity and pathway-response divergence. (A) All unique structure pairs, displayed as hexagonal bins with logarithmic count scaling. Dashed lines indicate Morgan Tanimoto similarity of 0.80 and the all-pair 95th percentile of global response distance. (B) An enlarged view of structurally eligible pairs. Orange points exceed a pathway-specific response threshold for at least one pathway.

For every pathway, the candidate-cliff frequency among structurally similar pairs was below the approximately 5% all-pair reference. Extreme response differences were therefore not de-

scriptively enriched among similar structures under the chosen criteria. Nevertheless, the identified pairs reveal localized exceptions to structural similarity that could complicate prediction. Their presence alone does not establish that activity cliffs explain the overall limited predictive performance, particularly because only 201 structures had a neighbor meeting the similarity threshold. These remain candidate cliffs: residual experimental variability and uncertainty in molecular representations may also contribute to the observed response differences.

Discussion and Limitations

This study evaluated whether molecular representations could predict compound-induced transcriptional pathway responses in A549 cells under a fixed treatment condition. Predictive performance was modest: the best-performing model, CheMeleon embeddings with pathway-specific radial basis function kernel ridge regression, achieved a mean test R^2 of 0.150 across 14 pathways. Performance nevertheless varied substantially between pathways, with test R^2 values of 0.361 for MAPK, 0.303 for EGFR, and 0.222 for p53, compared with 0.042 for Androgen. These findings indicate that the evaluated models captured some response variation associated with molecular structure, but substantial variation remained unexplained. The observed performance characterizes this dataset and modeling procedure; it does not establish an upper bound on the predictive information available from molecular structure.

The reliability analysis helps distinguish reproducibility of the response from its predictability using molecular features. Conditional cross-block ICC was positively associated with predictive performance across pathways, with a descriptive Spearman correlation of 0.86. This pattern is consistent with response reproducibility contributing to differences in prediction difficulty. However, relatively high reliability did not guarantee strong predictive performance. For example, Hypoxia had an ICC of 0.751 but a test R^2 of 0.108, indicating that reproducible differences between structures were only partially captured by the models. Experimental variability therefore cannot fully explain the limited performance. Moreover, the ICC estimates describe individual structure-block observations in the repeated-structure subset, whereas prediction targets were aggregated responses. They should consequently not be interpreted as directly applicable numerical ceilings for test R^2 .

Experimental heterogeneity remains an important limitation. Significant block-associated variation was detected in 10 of the 14 pathways, supporting adjustment for shared experimental shifts. Nevertheless, the additive adjustment assumes that a block produces the same pathway-specific shift for all structures measured within it. It cannot account for structure-by-block interactions, unequal measurement precision, or unrecorded experimental factors. Block-effect estimates also depend on the overlap of structures across blocks and may be less precise when this overlap is sparse. Because the adjusted responses were subsequently treated as fixed prediction targets, uncertainty in the estimated corrections was not propagated through the predictive analysis. Residual experimental variation could therefore contribute to prediction error, although the present results do not quantify its contribution or demonstrate that it dominates the remaining error.

The number and distribution of available structures may also have constrained model development. Although the curated dataset contained 2,355 signatures, these represented only 1,494 distinct structures. Repeated observations can improve response estimation but do not provide additional independent examples of molecular structure. Coverage of closely related structures was also limited: only 201 structures, approximately 13.5% of the dataset, had a neighbour meeting the Morgan similarity threshold of 0.80. Such sparse local coverage may make it difficult to learn how particular structural changes influence pathway responses. The

available sample size limits how reliably flexible models can be estimated and compared, although it does not preclude their use with suitable regularization or pretraining. The failure of the evaluated feedforward neural network to outperform the best kernel model suggests that increasing model flexibility alone was insufficient. Learning curves would help determine whether additional training structures are likely to produce substantial gains.

Structural curation reduced inconsistencies in the molecular inputs but could not eliminate uncertainty in the underlying annotations. Fifty of the 1,555 initially eligible compound identifiers lacked recoverable structures and were excluded. This exclusion reduced chemical coverage and could introduce selection bias if missing annotations were concentrated in particular compound classes. Among the retained compounds, 293 of 1,498 identifiers had unresolved stereochemical elements. Representations derived from these records cannot distinguish configurations that are absent from the input, potentially obscuring biologically relevant differences. However, unspecified stereochemistry does not necessarily indicate an erroneous annotation; it may reflect an unassigned configuration or a mixture. Similarly, recovery of a compatible PubChem structure does not independently verify the stereochemical composition of the material used in the experiment.

The representation comparisons suggest that predictive performance depends jointly on molecular features and the regression method. Morgan fingerprints performed better with ridge regression, whereas CheMeleon embeddings combined with pathway-specific nonlinear regression achieved the highest mean test R^2 . Thus, the results do not establish that either representation is uniformly superior. Because the nonlinear comparisons also involved different kernels and tuning procedures, they do not isolate representation quality alone. The frozen CheMeleon encoder was not adapted specifically to the pathway-response prediction task, which may limit the relevance of some learned features. Future comparisons could examine task-specific fine-tuning, additional physicochemical descriptors, or three-dimensional representations. However, these approaches require empirical evaluation: generated conformations may not capture biologically relevant configurations, and unresolved stereochemistry introduces additional uncertainty into three-dimensional inputs.

The biological distance between molecular structure and a downstream transcriptional response provides another plausible source of prediction difficulty. A pathway score measured after 24 hours may reflect multiple processes, including cellular exposure, target engagement, and subsequent responses or adaptation. Although cell line, nominal concentration, and treatment duration were held constant, these restrictions do not ensure equivalent intracellular exposure or target engagement across compounds. A single endpoint also cannot distinguish early responses from later consequences of treatment. These considerations motivate incorporating complementary biological information where available, but the present analysis does not identify which intermediate processes account for the observed errors. Furthermore, conclusions are restricted to A549 cells under the studied treatment condition and cannot be assumed to generalize to other cellular contexts, concentrations, or time points.

Uncertainty also arises from the definition of the prediction targets. The models predict PROGENy-derived transcriptional scores rather than directly measured biochemical pathway

activity. These scores depend on the selected response footprints and the expression values used to calculate them, including computationally inferred genes in the L1000 profiles. Sensitivity analyses examining footprint size, gene coverage, and the contribution of inferred genes could clarify how response construction affects reliability and prediction. Alternative scoring approaches or gene-set collections could extend the analysis to other biological processes. However, such alternatives would change the outcome being predicted, and a higher predictive R^2 would not by itself demonstrate a more biologically valid measurement.

The activity-cliff analysis provided limited evidence that widespread extreme response divergence among highly similar structures explained the modest overall performance. No global cliffs were identified among the 237 eligible pairs, although 23 pairs exceeded at least one pathway-specific threshold. These findings are conditional on the chosen fingerprint, similarity cutoff, and response-difference criteria. Limited close-neighbour coverage reduces the scope of the analysis, while aggregation across pathways can dilute differences confined to a single pathway. In addition, the analysis did not directly test whether structures involved in candidate cliffs had larger held-out prediction errors. Activity cliffs therefore remain a possible local challenge, but the current evidence does not support them as a principal explanation for overall prediction difficulty.

Finally, evaluation was based on a single structure-level split with 225 test structures. The observed model rankings and performance differences may therefore depend on the partition, and uncertainty estimates are needed before interpreting small differences as reliable advantages. Structure-level splitting prevents identical curated structures from appearing across partitions, but related analogues may still occur in both training and testing. Generalization to unfamiliar scaffolds, independent experimental campaigns, and external datasets remains untested. The pooled reliability and activity-cliff analyses provide descriptive context rather than independent validation of predictive performance. Future work should prioritize repeated or nested evaluation, learning curves, and external or scaffold-based testing, alongside targeted sensitivity analyses of structural annotations and pathway-score construction. These steps would help identify which limitations are most consequential and whether improvements translate into more reliable predictions. Functional validation would additionally be necessary before interpreting predicted pathway responses as evidence of anticancer efficacy.

References

- ATCC. n.d. *A549 (CCL-185): Human Lung Carcinoma Cell Line*. <https://www.atcc.org/products/ccl-185>.
- Badia-i-Mompel, Pau et al. 2022. “decoupleR: Ensemble of Computational Methods to Infer Biological Activities from Omics Data.” *Bioinformatics Advances* 2 (1): vbac016. <https://doi.org/10.1093/bioadv/vbac016>.
- Benjamini, Yoav, and Yosef Hochberg. 1995. “Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing.” *Journal of the Royal Statistical Society: Series B* 57 (1): 289–300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>.
- Bento, A. Patrícia, Anne Hersey, Eloy Félix, et al. 2020. “An Open Source Chemical Structure Curation Pipeline Using RDKit.” *Journal of Cheminformatics* 12: 51. <https://doi.org/10.1186/s13321-020-00456-1>.
- Burns, Jackson W., Akshat Shirish Zalte, Charles R. A. Abreu, et al. 2026. *Deep Learning Foundation Models from Classical Molecular Descriptors*. <https://arxiv.org/abs/2506.15792>.
- Burns, Jackson, Akshat Zalte, and William Green. 2025. *Descriptor-Based Foundation Models for Molecular Property Prediction*. <https://arxiv.org/abs/2506.15792v1>.
- Cherkasov, Artem, Eugene N. Muratov, Denis Fourches, et al. 2014. “QSAR Modeling: Where Have You Been? Where Are You Going To?” *Journal of Medicinal Chemistry* 57 (12): 4977–5010. <https://doi.org/10.1021/jm4004285>.
- Drucker, Harris, Christopher J. C. Burges, Linda Kaufman, Alex J. Smola, and Vladimir Vapnik. 1997. “Support Vector Regression Machines.” In *Advances in Neural Information Processing Systems 9*, edited by Michael C. Mozer, Michael I. Jordan, and Thomas Petsche. MIT Press. https://proceedings.neurips.cc/paper_files/paper/1996/hash/d38901788c533e8286cb6400b40b386d-Abstract.html.
- Giard, Donald J., Stuart A. Aaronson, George J. Todaro, et al. 1973. “In Vitro Cultivation of Human Tumors: Establishment of Cell Lines Derived from a Series of Solid Tumors.” *Journal of the National Cancer Institute* 51 (5): 1417–23. <https://doi.org/10.1093/jnci/51.5.1417>.
- Hoerl, Arthur E., and Robert W. Kennard. 1970. “Ridge Regression: Biased Estimation for Nonorthogonal Problems.” *Technometrics* 12 (1): 55–67. <https://doi.org/10.1080/00401706.1970.10488634>.

- Holland, Christian H. et al. 2020. “Robustness and Applicability of Transcription Factor and Pathway Analysis Tools on Single-Cell RNA-Seq Data.” *Genome Biology* 21: 36. <https://doi.org/10.1186/s13059-020-1949-z>.
- Kim, Sunghwan, Jie Chen, Tiejun Cheng, et al. 2025. “PubChem 2025 Update.” *Nucleic Acids Research* 53 (D1): D1516–25. <https://doi.org/10.1093/nar/gkae1059>.
- Paszke, Adam, Sam Gross, Francisco Massa, et al. 2019. “PyTorch: An Imperative Style, High-Performance Deep Learning Library.” *Advances in Neural Information Processing Systems* 32. https://proceedings.neurips.cc/paper_files/paper/2019/hash/bdbca288fee7f92f2bfa9f7012727740-Abstract.html.
- RDKit contributors. 2026. *RDKit: Open-Source Cheminformatics Software*. V. 2026.03.5. Released. <https://www.rdkit.org/>.
- Rogers, David, and Mathew Hahn. 2010. “Extended-Connectivity Fingerprints.” *Journal of Chemical Information and Modeling* 50 (5): 742–54. <https://doi.org/10.1021/ci100050t>.
- Saez Lab. n.d. *PROGENy: Pathway Responsive GENes for Activity Inference from Gene Expression*. Accessed September 15, 2026. <https://saezlab.github.io/progeny/>.
- Schubert, Michael, Bertram Klinger, Martina Klünemann, et al. 2018. “Perturbation-Response Genes Reveal Signaling Footprints in Cancer Gene Expression.” *Nature Communications* 9: 20. <https://doi.org/10.1038/s41467-017-02391-6>.
- Subramanian, Aravind et al. 2017. “A Next Generation Connectivity Map: L1000 Platform and the First 1,000,000 Profiles.” *Cell* 171 (6): 1437–1452.e17. <https://doi.org/10.1016/j.cell.2017.10.049>.
- The Cancer Genome Atlas Research Network. 2014. “Comprehensive Molecular Profiling of Lung Adenocarcinoma.” *Nature* 511 (7511): 543–50. <https://doi.org/10.1038/nature13385>.
- Tilborg, Derek van, Alisa Alenicheva, and Francesca Grisoni. 2022. “Exposing the Limitations of Molecular Machine Learning with Activity Cliffs.” *Journal of Chemical Information and Modeling* 62 (23): 5938–51. <https://doi.org/10.1021/acs.jcim.2c01073>.